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(Key and Solutions for AIMCAT2406)

Key

SECTION – I

1. A	7. B	13. A	19. C
2. C	8. C	14. A	20. 4132
3. A	9. A	15. C	21. 4312
4. B	10. C	16. C	22. C
5. D	11. A	17. B	23. 4213
6. C	12. D	18. B	24. A

SECTION – II

1. 468	6. A	11. A	16. 26
2. A	7. B	12. 50	17. C
3. 87814080	8. D	13. A	18. 0
4. C	9. C	14. 20	19. B
5. D	10. B	15. D	20. C

SECTION – III

1. 384	7. B	13. D	19. D
2. 7	8. 8	14. C	20. 270
3. 11	9. B	15. A	21. C
4. C	10. B	16. B	22. C
5. 17	11. A	17. 696	
6. D	12. D	18. 14520	

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 535

1. Option A: In the passage, the author uses the analogy of osmosis to better explain the immigration crisis and the causes for this crisis. The author believes that policymakers do not understand that migration is a natural process, similar to osmosis. This means that if they were to understand this, then they would take measures to address the underlying issues that cause migration rather than persecute migration itself. This is one of the author's suggestions.
Option B: From the lines "population flow from nation to nation will be a direct function of the degree of difference of goods, opportunities and hope" and "…but without a dramatic redistribution of goods and opportunities… we can expect worldwide population shifts to only increase", we understand that the author recommends redistribution of goods and services. However, the author is of the opinion that it should reduce the gradients across countries (only then the problem of immigration can be addressed). This option mentions redistribution within a country which may or may not be effective. This cannot be inferred.
Option C: The author suggests accepting the causes of migration as natural and unavoidable, which means that migration in itself should be considered as the same. However, the author does not suggest that the solution to this is to open borders of nations to migrants. He suggests people having equal access to opportunities rather than the nature of permeability or impermeability of borders. Hence, this also cannot be inferred.
Option D: The author talks about climate change and population explosion in poor countries as major catalysts for the immigration crisis. However, he does not recommend that we should address these issues. He only recommends that, while formulating policies for immigration, these factors must be considered by the policy makers. Hence, Option D is not the answer.
Therefore, Option A is the correct answer.
Choice (A)

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2. The author uses the line "When a patient has a serious illness confirmed, it is futile to keep repeating the lab tests in hopes of a different result". Here, the serious illness can be equated to the migration crisis and the lab tests giving the results would allude to the data that points to or proves that there is a migration crisis. The author mentions that policymakers (those in charge) choose to focus on cherry-picking data such as where and to what extent there is population expansion and climate change – however, this is an exercise in futility since neither does it negate the fact that there is a migration crisis, nor does it do anything to solve the problems that are causing the migration crisis.
Option A: It is incorrect to say that policymakers deny the existence of immigration crisis. They divert attention from it. Hence, this is incorrect.
Option B: This option does talk about policymakers cherry-picking data; it also states that they do this to "divert attention from the necessary measures needed to deal with the crisis of worldwide population shifts". This is too strong a statement to be gleaned from the author's ideas. This is not the correct answer.
Option C: Arguing about factors such as "degree and maximal location of upcoming population expansion and climate change" does not negate the fact that there is a migration crisis caused by population expansion and climate change. Highlighting information that shows the crisis in a better light does not alleviate the problem. This is what can be inferred from the given sentence. This is the correct answer.
Option D: While the measures necessary to deal with the problems of population expansion and climate change are immensely harsh and those in charge do not like implementing them, we cannot infer that they are looking for other measures that are not so draconian. Further, this is unrelated to the sentence given in the question. This is not the answer.
Choice (C)

3. The author mentions that policymakers perpetuate the imbalances that cause migration by closing a nation's borders and increasing security at the border. The author however believes that these obstacles are not impossible to overcome for someone who is absolutely desperate and has no choice but to migrate to a new, better place. The author is likely to quote something that emphasizes this train of thought.
Option A: Home is the mouth of a shark implies that home is a place that is dangerous. This means that no one would want to leave their home unless forced to or unless they are desperate to escape a bad situation. This is continuing that author's line of thought that migrants are undeterred by the obstacles they face because they are desperate to get away from the place they call home. Option A is the correct answer.
Option B: This option likely refers to the actions of policymakers who decide/in the past, decided the boundaries of nations and in the process acquired ownership of goods and opportunities not available to others. This talks about the author's idea that policymakers perpetuate imbalances by building walls and borders. But the author discontinues this train of thought and talks about the desperation that migrants face, that pushes them to leave home. This is not something the author will add at the end of the 5th para and so, it is not the answer.
Option C: Here, the word "we" refers to migrants. The author doesn't attribute immigration just to wars. Hence, Option C is not the answer.
Option D: This option does in no way allude to the desperation that migrants face. It is more of a philosophical idea, where the mind is free to travel and has no boundaries and so, the poet calls the mind "my immigrant mind". This is not the answer.
Choice (A)

4. Option A: In the passage the author mentions that "the most popular and immediate reaction is to decrease the permeability of the membranes separating countries" and planning on erecting theoretically insurmountable walls, is not a logical solution. This option talks about how the Department of Homeland Security is planning to build a new border wall. The author is not likely to agree with this.
Option B: This option talks about how the EU is planning on having a unified asylum policy (for all the countries in the EU council) and having a better management of migrant flows, which could actually support population flows and make lives easier for migrants. This does not talk about keeping out migrants, tightening border control, and other such strategies that the author does not approve of. This is the correct answer.
Option C: The author states that "…shift of populations isn't just an ethical or metaphysical dilemma to be resolved at the level of 'us' versus 'them'. It isn't about the right to own land and enforce borders, or the relative worth of individuals versus groups." This option is about closing borders to address the migrant issue at a "ethical/metaphysical" level. The author will not endorse this.
Option D: Similar to Option C, this showcases a strategy about dealing with migrants at a 'us' versus 'them' level. It talks about the relative worth of a group – migrants being considered as terrorists. Stopping migrants before they reach the borders does nothing to help migrants or relieve them of the issues they are facing. This is not the answer since the author would not approve of this.
Choice (B)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 548

5. Option A: The author says nothing about the difficulty as far as comprehension of the process is concerned. In fact, the author doesn't discuss the process of decomposition anywhere in the passage. Option A is not the answer.

Option B: Given that the author starts the line with 'As with many green practices' it can be understood that whatever the author is talking about, applies to all green practices. It is certainly not being conveyed that this is not a green practice. Hence, Option B is not the answer.
Option C: The efficiency of the process of decomposition is something the author doesn't discuss. Neither does the passage discuss human intervention in the process of decomposition. Option C is not the answer.
Option D: The 'devil is in the details' is an idiomatic expression which means that once we look deeper there will be facts that prove our assumptions wrong. There will be a fine print. In this case, while it is assumed that biodegradable plastic will decompose once it is dumped, it cannot be thought to be so. There is fine print involved, and hence, it cannot be a foregone conclusion. Option D is the answer.
Choice (D)

6. Option A: We are talking about biodegradable plastic as the silver-bullet solution here. The author does mention that green initiatives are a bit complex to comprehend, but doesn't call them a hoax outright. Also, the question is specifically about the silver bullet solution. Hence, Option A is not the answer.
Option B: The author mentions that the idea of recycling may not always be environment friendly if we spend so many resources to make these materials in the first place. However, the author doesn't particularly mention that biodegradable plastic is not environment friendly. Hence, Option B is not the answer.
Option C: The silver bullet solution here is that of using biodegradable plastic. The author's main argument against this is that it is a misunderstood term and it doesn't work the way people think it does. In simple terms, the plastic is 'biodegradable' but with fine print and with certain caveats. These caveats make the process tricky unlike what consumers of biodegradable plastic imagine. Option C is the answer.
Option D: The author doesn't really go all out and assert that biodegradable plastic doesn't decompose at all (if that were the case, municipal green waste collection systems wouldn't have been mentioned at all). The author simply indicates that it is not as simple as the term suggests. Option D is not the answer.
Choice (C)

7. Option A: This is true as the author says that only 5% consumers have compost piles and the temperature doesn't get high enough even in those to decompose PLA. Hence, Option A is not the answer.
Option B: This is incorrect. The author mentions that recycling biodegradable plastic along with normally recycled material affects the quality of the 'plastic' that is made from that material. That plastic manufactured from recyclable material may or may not be 'recyclable plastic'. The passage doesn't talk about the recyclability of the plastic made from biodegradable material. Also, biodegradable material lowers the quality of the plastic made from other recyclable materials, not just from itself. Hence, Option B is the answer.
Option C: This is true, as the author says that green waste collection systems are only present in a few zip codes and that limits the extent to which biodegradable plastic can really be gotten rid of or disposed of. Option C is not the answer.
Option D: This has been mentioned in the passage. Biodegradable plastic doesn't decompose well in landfills because they are a little low on oxygen. So, throwing biodegradable cups in a normal trash bin is as good as using normal plastic and disposing it. Hence, Option D is not the answer.
Choice (B)

8. The author's argument is to conserve energy by not trying to decompose plastic, and instead use it for as long as possible or recycle it and use it.
Option A: This doesn't weaken the author's argument because plastic not decomposing doesn't really tell us why we have to decompose it in the first place at all (which is what the author suggests too – that we need not decompose the plastic). Option A is not the answer.
Option B: If they cannot be used, they can be recycled. That is the author's argument. His argument is against decomposition, not recycling. Hence, Option B doesn't weaken his answer.
Option C: This weakens the author's argument because if biodegradable plastic is more energy-efficient to make, it goes against the author's suggestion to keep plastic as plastic for as long as possible. This option also further weakens the author's argument because of the mention of longer shelf life of biodegradable plastic – so there is no way normal plastic can be more energy-efficient as the author suggested in the passage. Hence, Option C is the answer.
Option D: The author's argument is in favour of keeping plastic as plastic for as long as possible. So, even if decomposition doesn't cost energy, it doesn't necessarily make a case against the author's suggestion. That is because the author believes it is a waste of energy to decompose plastic (irrespective of whether decomposition is energy-efficient or not) because a lot of energy has been used to make that plastic. Option D is not the answer.
Choice (C)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 554

9. Option A: The para starts with a cause - water at the surface, and then moves on to discuss water infiltration and pressure change propagation and then ends with rock failure at depths. So, two things can be understood from this, that fluid pressure propagation corresponds to rock failure (as understood from 'downward migration of rock

failure over time) and that water (the fluid here) infiltrating has a connection to the pressure propagating downward (hence, the time lag that percolation takes). So, Option A is the answer.
Option B: The passage mentions 'a time lag between the accumulation of water at the surface and failure at depth', which means we cannot be sure that the water percolates to the depths (completely). But more importantly, we cannot infer that it is 'difficult' to dismantle the rocks. That understanding cannot lead to the theory mentioned in the question. Option B is not the answer.
Option C: It is highly inaccurate to say that the water accumulated on the surface is what influences fluid pressure changes. We can understand from the para that it is more to do with the water infiltrating into the permeable rocks. Hence, Option C can be eliminated.
Option D: The para does talk about a time lag between water accumulated at the surface and rock failure at the bottom. The para doesn't really discuss time taken for the actual water (and not the rock failure) to propagate to the depths. So, it is inaccurate, and more importantly something that is not in sync with the theory. Option D is not the answer.
Choice (A)

10. Option A: These eruptions could have taken place anywhere, for example, in the upper rift zone. This statement, even if true, doesn't throw a negative light on the hypothesis that rainfall was responsible for the eruption on the lower east rift zone. Only if eruptions in the lower east rift zone take place in the dry season do we need to reconsider the hypothesis. Option A is not the answer.
Option B: The passage clearly says that sometimes the rock is weak/there are faults, and rainfall only exacerbates the situation, which still means their proposed theory is correct. So, if this statement is true, it doesn't take away anything from their argument. Option B is not the answer.
Option C: This proposes a completely different hypothesis to explain the eruption on the lower east rift zone. If this is true, then the rainfall need not have contributed at all to the eruption. That will not invalidate their theory but will definitely weaken the theory. Option C is the answer.
Option D: Even if the pressure differential varies from one rainfall to another, it still means that there is a pressure differential thus proving the theory that rainfall does alter the pressure (positively) at a depth of several kilometres. This option doesn't weaken the argument.
Choice (C)

11. Option A: This will be able to answer the million-dollar question. Water percolation can lead to rock failure as established in the passage conclusively. Can that lead directly to an eruption or to an earthquake which leads to an eruption is what the uncertainty is about. So, from the last line, we can understand that rock failure close to magma is what we are trying to observe and if water failure causes it then we can make a decisive argument. Option A is the answer.
Option B: The study by 'Farquharson and Amelung' has already done this showing that water-pressure changes at great depths correspond to rock failure. Whether that would lead to a volcanic eruption is the unanswered question in the passage, as mentioned in the last line. We are uncertain about whether water-pressure changes can lead to rock failure around the point where magma is stored leading to eruptions. Option B is not the answer.
Option C: The last para clearly mentions that for an eruption to take place magma has to generate a pathway through the rocks (something that becomes difficult if there is an earthquake that releases the stress in the rocks). So, this study will not add a lot of value to the discussion in the passage, since the last sentence questions failure in rocks caused by water pressure changes *around magma storage*. Option C is not the answer.
Option D: The passage establishes that there is a correlation between rainfall and earthquakes as understood from 'Although it is well established that changes in water pressure promote earthquakes…'. However, the passage is ambiguous about the correlation/causation between eruptions and earthquakes mentioning both (In favour of a correlation - *examples of earthquakes caused by pressure changes on this scale are abundant* and Against a correlation - *as such, they might make it more difficult for magma to erupt*). So, this study will help answer part of the question. However, the main hypothesis is about whether water failure causes eruptions or eventually leads to eruptions. This study can still not throw light on that. Option D is close but not the answer.
Choice (A)

12. Option A: The para doesn't indulge in such theories at all, and doesn't even try to establish causation for earthquakes (we are trying to establish causations for eruptions, through most of the passage). So, this example is definitely not to show (since we cannot generalize) that eruptions lead to earthquakes. Option A is not the answer.
Option B: This is true, but is stating the obvious. The author took the example of an event where pressure change led to an earthquake and that was to explain that pressure changes do cause rock failures. So, Option B is true but not the purpose of why the example was mentioned (an example is mentioned to prove a higher point, not to prove the example itself). Option B is not the answer.
Option C: The para doesn't really distinguish between rocks close to breaking point and rocks not so close to breaking point. All we are discussing is water pressure changes not being so high, but still a cause of failures in rocks when the rocks are already on the verge of a failure. Option C is not the answer.

Option D: The para mentions that '*if rocks are already close to breaking, such changes might be sufficient to initiate failure*', and then goes on to mention the earthquake caused by the eruption. This is followed by the remark that many earthquakes are caused by pressure changes on this scale, leading us to connect earthquakes and rock failure, the latter being caused by water pressure changes. Option D is the answer.
Choice (D)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 505

13. "In a blow to gym vloggers who pretend to be scientists everywhere, the paleo diet may not be all that paleo. According to a new study, humans were eating carbs up to 170,000 years ago." It can be inferred from the passage that paleo diet is the diet followed by our ancestors thousands of years ago and that it refrains from carbs. The new study reveals that it was not the case with our ancestors, who used to consume carbs. It would be a blow to gym vloggers if this information will affect them negatively.
Choice (A): If their only shot to fame is the information they share, the information revealed by the new study would be a blow to them because it is in contradiction with what proponents of the paleo diet say. Therefore, this new piece of information would be a blow to them as they might lose their credibility and subsequently, their fame. Therefore, this would strengthen the author's statement. Hence, this is the answer.
Choice (B): Spreading the word about paleo diet need not necessarily mean endorsing paleo diet. If gym vloggers urge people to refrain from paleo diet, they would still be spreading the word about paleo diet. Therefore, this need not be a blow to them. So, this need not strengthen the author's statement. Hence, this is not the answer.
Choice (C): Gym vloggers can still maintain their bodies despite the information revealed by the new study by adapting to it accordingly. Therefore, this need not be a blow. So this need not strengthen the author's statement. Hence, this is not the answer.
Choice (D): They might simply add the new piece of information to their knowledge or they can still pretend that they know everything even if that is not true. Either of these is not a blow to them. So this need not strengthen the author's statement. Hence, this is not the answer.
Choice (A)

14. The author mentions "African potato" in the third paragraph of the passage.
Choice (A): In the third paragraph, the author asks the reader if the latter wonders whether a starchy plant is really a carb and as an answer to that, the author goes on to give the example of what is nicknamed as "African potato." Here, the tone of the author is in the affirmative. And it is not the 'African' in the name that confirms that, since the study the author mentions concerns Africans. It is the 'potato' in the name that confirms the presence of carbs. From this, it can be understood that the assumption here is that a plant that has 'potato' in its name has carbs in it. Hence, this is the answer.
Choice (B): Although this can be inferred from the third paragraph of the passage that the African potato is starchy, this is not an assumption that the author makes. This is specific to only the particular plant that the author refers to. Therefore, it cannot be said that the author assumes that any other plant that is named so must be starchy. Hence, this is not the answer.
Choice (C): Here, the author is asking about a starchy plant and not the charred remnants of a plant. Therefore, the assumption, if any, that the author made has to be regarding a plant and not the charred remnants of a plant. Therefore, this need not be the assumption that the author made. Hence, this is not the answer.
Choice (D): It has African in its name because the ancestors that the author refers to in the previous statement are Africans. Therefore, this need not be the assumption that the author made. Hence, this is not the answer.
Choice (A)

15. Choice (A): It is stated in the fourth paragraph that there is evidence that humans did not necessarily evolve to eat meat. Whether the proponents of paleo diet believe it or not cannot be inferred. Hence, this is not the answer.
Choice (B): It is not mentioned in the passage as to what the proponents of the paleo diet do with any evidence that points towards paleo diet consisting of starchy grains. Hence, this is not the answer.
Choice (C): It is stated that the proponents of paleo diet advocate refraining from carbs and endorse the idea of eating like our ancestors. In the second paragraph, it is stated that carbohydrates found in grains are cut out from the paleo diet. But the author mentions a study in the third paragraph as evidence suggesting that our ancestors did consume carbs found in grains. Therefore, it can be understood that the proponents of paleo diet are under the false impression that the diet of our ancestors was devoid of carbohydrates found in grains. Hence, this is the answer.
Choice (D): Although the proponents of the paleo diet consider their ideology as theoretical in nature, there is no information in the passage regarding why they considered this. Therefore, this cannot be inferred to be true. Hence, this is not the answer.
Choice (C)

16. "Seeing as potatoes were first domesticated between 8000 and 5000 BC, this seems like bad science – but then again, what about basing a modern diet on the proposed practices of cavemen doesn't?"
Choice (A): The author mentions "bad science" while discussing paleo diet, not potatoes. Hence, this is not the answer.

Choice (B): The author concludes the passage with a question that implies that nothing about basing a modern diet on the **proposed practices** of cavemen does not seem like bad science. If those proposed habits are healthy, then it need not be a case of bad science because the idea of maintaining any kind of diet is to be healthy. Therefore, in this case, it need not be bad science. Hence, this is not the answer.
Choice (C): Some proponents of paleo diet, who endorse the diet of our ancestors, say that potatoes are forbidden based on '***Paleo is all about eating only those foods that people living the pre-agricultural age could have eaten***.' But the fact that potatoes were first domesticated between 8000 and 5000 BC suggests that our ancestors were consuming potatoes and thus, the notion of some of those proponents fails logic. Therefore, this seems like bad science. Hence, this is the answer.
Choice (D): The difference in the diets of modern and ancient humans is not what is bad science. The practice of basing modern diet on ancient practices is what the author refers to as bad science. Hence, this is not the answer.
Choice (C)

Solutions for questions 17 and 18:

17. Option A: The para is about empathy and the two studies/surveys are only to further the author's point. The essence of the para is about the importance of empathy, and not about salaries and grievances in particular which are just two highly specific examples. Option A is not the answer.
Option B: This option offers both the important ideas in the para: the importance of empathy to collaboration, and its importance particularly in a global economy. Option B is the answer.
Option C: There are two problems with this option. The para is not just about the relationship between empathy and collaboration (as opposed to empathy in a collaborative enterprise). The other problem is the reverse conclusion. Empathy is important to collaboration, but the para doesn't indicate that collaboration is impossible to empathy. Option C is not the answer.
Option D: While we can read the ability to interpret and to respond to those different from us as empathy, the essence of the para is about empathy being important for collaboration, and how emotions affect the bottomline. This option states that empathy affects the bottomline. Option D is not the answer.
Choice (B)

Solution for question 18:

18. Option A: Museums do not remove works of art from the places they were originally present in with the goal to 'rescue them.' The para indicates that by collecting all the works of art, they metaphorically allow the past to survive (in the museum). That doesn't mean that the works of art were all necessarily in danger. Option A is not the answer.
Option B: Historians disapprove of museums. That covers one part of the idea. Museums are still important. That is the second part of the idea. They are critical because they allow at least some part of the past to survive. That is the third part of the idea. This option summarises all the ideas of the para. Option B is the answer.
Option C: Firstly, it is incorrect to say that museums deform the past. They probably deform the works of art according to the author. Secondly, the author doesn't call them an evil. The author thinks they are useful. Hence, this option can be easily eliminated. Option C is not the answer.
Option D: The para doesn't differentiate between two groups of works of art, where one group is rescued and the other deformed. The author talks about all works of art as one group, whose essence is sometimes erased by museums, although he/she states that this is one way the past can survive at least. Option D is not the answer.
Choice (B)

Solution for question 19:

19. The "though" in the highlighted sentence implies that it should contrast a prior idea. The highlighted sentence states that something was not perceived as a serious moral issue. So the statement prior to the highlighted sentence should state that something was perceived as a serious moral issue.
The statements immediately preceding blank (1) and blank (2) both do not mention anything that is seen as a moral issue. Therefore, the highlighted sentence has to follow these statements. Hence, the highlighted sentence does not fit in blank (1) and blank (2).
The statement preceding blank (3) states that phrenology attempted to pinpoint the location of different mental functions in different parts of the brain. It states that it was seen as heretical, a serious moral issue. But doing something similar – discover a person's character and intellect based on the size and shape of their head was not perceived as a serious moral issue. Even though both processes attempt to do the same, one is seen as heretical and one is seen as uncontroversial. This is the contrast between the highlighted sentence and the statement preceding blank (3). Therefore, these are directly connected. Hence, the highlighted sentence fits in blank (3).
The statement preceding blank (4) is along the lines of the highlighted sentence. But the highlighted sentence needs a contrasting statement preceding it (as inferred from the "though" in the highlighted sentence). Hence, the highlighted sentence does not fit in blank (4).
Choice (C)

Solutions for questions 20 and 21:

20. Sentence 1 references a previous enquiry with 'such' (we have to look for questions in other sentences).
Sentence 2 starts with a positive connector 'As a result', and explains the consequences of the internet misleading us due to our bias.
Sentence 3 is an independent sentence about the Internet, explaining why people separate into groups.
Sentence 4 enquires about the consequences of the collective will of the majority.
We can easily combine 4 and 1, 4 enquiring and 1 pointing to the enquiry. Similarly, 3 and 2 can be clubbed together with 2 elaborating on the consequences of the way the Internet operate.
The only tricky bit is to arrange the 41 and 32 blocks. The 32 block cannot precede the 41 block, because 4 starts with a question that includes both positive and negative possibilities (democratic consensus or fundamentalist groupthink). So, 4132.
Ans: (4132)

21. Sentence 1 uses 'it' to point to something that speaks to our feelings and imaginations.
Sentence 2 starts with a positive connector and then explains how instrumental music is different from prose and poetry.
Sentence 3 is about music being different from other static arts (it should be noted that music is a bigger idea than instrumental music).
Sentence 4 says we can get insights by comparing music with other familiar arts. It doesn't mention what the insight is.
We need to understand that 43 is a block. 4 comes before 3 because of the way the ideas have been presented. It starts with why we are talking about music – that we are looking for insights into its nature. 4 is also the sentence that in the second part discusses a comparison, which is elaborated in 3 – the difference between music and the static arts. Obviously, 1 and 2 come after the 43 block because both are dependent. 1 comes with 'it' and 2 comes with 'Furthermore'. 1 must come after 3 (and not after 2) because 1 is not about instrumental music but about music in general. This can be understood from the fact that 2 moves to language, unlike in 3 which talks about all static arts. Also, 'intangible' in 3 points to what 1 says – speaking to our feelings and imagination. So, 4312 is the order.
Ans: (4312)

Solution for question 22:

22. Note: Philology is the branch of knowledge that deals with the structure, development and relationships of a language.
The statement immediately preceding blank (1) talks about Bahar's civilisational paradigm coming from European philology while the statement immediately following blank (1) talks about how his efforts were shaped by European Orientalism. Therefore, these two statements are connected and the highlighted sentence, which talks about New Persian, will break the flow. Hence, it cannot fit into blank (1).
The statement immediately following blank (2) has the phrase "These European philologists". "These" here refers to proponents of European Orientalism, mentioned in the statement immediately preceding blank (2). Hence, the two statements are connected and inserting the highlighted sentence in blank (2) would break the flow. Hence, the highlighted sentence does not fit into blank (2).
The statement immediately preceding blank (3) introduces the Persian language. The highlighted sentence talks about New Persian. "they" in the highlighted sentence refers to the early 19th-century philologists, Bahar and many others mentioned in the statement immediately preceding blank (1). Hence, these two statements are connected. Besides, in the statement immediately following blank (3), "in doing so" refers to "built a narrative that emphasized…" in the highlighted sentence. Hence the highlighted sentence fits into blank (3).
The statement immediately preceding blank (4) talks about how 19th century German linguists classified 'Old English' and 'Middle English' as precursors to modern English. This is in line with what is mentioned in the highlighted sentence. However, as this statement begins with "in doing so", it can be inferred that it parallels what is mentioned in the highlighted sentence. Therefore, the highlighted sentence has to precede this statement. Hence, the highlighted sentence does not fit into blank (4).
Choice (C)

Solution for question 23:

23. Sentence 1 is a dependent sentence due the positive connector 'also' referencing a previous sentence where trust and expertise have been discussed as being important.
Sentence 2 is again a dependent sentence with a positive connection starting with 'for example' – this sentence must follow a sentence that emphasizes upon the importance of trust.
Sentence 3 refers to a previous subject with 'Here', somewhere where trust and expertise are important.
Sentence 4 is an independent sentence establish the importance of trust and expertise in AI systems.
4 is the only independent sentence establishing the context. So, it will begin the para. Similarly, 1 is downstream to 4 as a different scenario where trust and expertise is 'also' important. That leaves us with 2 and 3. 2 is an example for 4 because it talks about information search systems, referencing the interest for computer scientists in 4. 3 is the elaboration of 1 discussing argumentation. Therefore, 4213 is the order.
Ans: (4213)

Solution for question 24:

24. Option A: This option mentions all sub-ideas, that it is Hobbesian thought which delineates ethics in politics from ethics in our personal associations. It also mentions how ethics plays a greater/more effective role in personal relationships. Option A is the answer.
Option B: While this is true, in the general sense, that the role of morality is different in the personal and collective dominions, the option doesn't indicate an important idea which is that the regulation of collective actions by morality is not as much as it is in personal relationships. Morality is more effective in personal relationships. Option B is not the answer.
Option C: The para mentions that politics cannot be straightforwardly regulated by morality. However, it cannot be summarised to say that politics and morality don't go hand in hand at all. Option C is not the answer.
Option D: The para does state that politics is precisely necessary to resolve the moral disagreements, and to regulate our collective action by deciding the extent to which moral values underpin or regulate that collective action. However, this option in itself doesn't represent the essence of the para because the main idea is a comparison of the collective principles as opposed to those guiding personal associations. The mention of the distinction between ethics in personal and political associations has been omitted. Option D is not the answer.
Choice (A)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	5, 6, 7, 13, 15, 16, 19, 20, 23
Difficult	1, 2, 3, 4, 8, 9, 11, 12, 14, 17, 18, 22, 24
Very Difficult	10, 21

SECTION – II

Solutions for questions 1 to 5:

From (iv), the sixth digit of N and the second digit of M are both 4. From (vii), the third digit of M and the fifth digit of N are both 5.
From (i), the first digit of N must be 1 (if it is more than 1, then seven times this digit will not be a single digit). The first digit of M must be 7 or 8.
From (iii), the fourth digit of M must be 1 or 2 or 3 or 4. Only then the seventh digit of N can be a single digit. The seventh digit of N can be 2 or 4 or 6 or 8. However, the seventh digit of N cannot be 4 (since the sixth digit of N is 4). Hence, the fourth digit of M cannot be 2. Further, the fourth digit of M cannot be 4 (since the second digit of M is 4). Hence, the seventh digit of N cannot be 8.
If the fourth digit of M is 1, from (iii), the third digit of N is 0. This is not possible as 0 cannot be one of the digits.
The fourth digit of M can only be 3. The seventh digit of N can only be 6 and the third digit of N must be 2.
From (v), the eighth digits of M and N can be 1 and 7 or 2 and 8 (in any order).
The eighth digit of N cannot be 1 or 2 (as they are the first and third digits of N). Hence, the eighth digit of N must be 7 or 8. The eighth digit of M must be 1 or 2.
From (ii), the fifth digit of M must also be 7 or 8. From (vi), the second digit of N must be 2 or 3. However, the second digit of N cannot be 2 (as it is the third digit of N). Hence, the second digit of N must be 3. The fifth digit of M must be 8. The eighth digit of N must also be 8. The eighth digit of M must be 2 and the first digit of M must be 7.
The fourth digit of N must be 7 (as it is the only digit remaining).
The seventh digit of M cannot be 6 (from (viii)). Hence, the seventh digit of M must be 1 and the sixth digit of M must be 6.
Hence, N = 13275468 and M = 74538612.

1. The number formed by the sixth, seventh and eighth digits of N, taken in that order, is 468.
Ans: (468)

2. The difference between the second digit of N and the second digit of M = 1
Choice (A)

3. The sum of M and N = 87814080
Ans: (87814080)

4. The sum of the fourth digit of N and the sixth digit of M = 7 + 6 = 13
Choice (C)

5. The first digit of N is 1. This is the same as the seventh digit of M.
Choice (D)

Solutions for questions 6 to 10:

From the given table, we can see that P1 and P5 depend on P3, P3 depends on P2, P2 depends on P4, P4 depends on P6 and P6 depends on P4. We can start with either P1 or P5 and find the minimum and maximum times at which each process can be finished.

6. Let the time at which P5 starts be 0th minute. Since P5 takes 27 minutes to finish, P5 will end at 27th minute.
From the table, we can observe that P5 is linked to P3.
Given that P5 must end before P3 ends. Also P5 must end at most 25 minutes before P3 ends. To minimize the total time taken by P5 and P3, we can consider that P3 ends at 28th minute. In this case, P3 must start at 9th minute (since P3 requires 19 minutes to complete). P1 and P2 both depend on P3. P1 must start after P3 ends. To minimize the time, the earliest that P1 can start is at the 29th minute. Hence, P1 will end by the 57th minute. P3 must end exactly 10 minutes before P2 ends. Hence, P2 must end at the 38th minute (since P3 ends at the 28th minute). P2 must start at the 6th minute.

P2 must start at most 15 minutes before P4 ends. The latest that P4 can end is at the 21st minute (since P2 starts at the 6th minute). The earliest that P4 can end is at the 14th minute, if P4 starts at the 0th minute (we need not take any earlier than 0th minute as the total time for all the processes will increase because of this).
P4 must start exactly 20 minutes after P6 starts. Hence, if P4 starts at the 0th minute, P6 must start at -20th minute. In this case, P6 will end at the 13th minute.
If P4 starts at 7th minute, P6 must start at -13th minute. In this case, P6 will end at the 20th minute.
In either case, the latest time which all the six processes will be complete is at the end of the 57th minute (for P1).
In either of the case, P6 starts at least 15 minutes before P4 ends. Hence, both these cases are valid.
To minimize the total time taken for all the processes, we need to find the latest time at which any process starts. P6 will be the earliest process and to minimize the total time, we can take P6 to start at -13th minute and end at 20th minute.
If we add 13 to all the starting times and ending times of all the processes, we get the following table on when each process should start:

Process	Start Time	End Time
P5	13	40
P3	22	41
P2	19	51
P1	42	70
P4	20	34
P6	0	33

Now, if we consider P5, the only constraint on P5 is that it must end before P3 ends and must end at most 25 minutes before P3 ends. Hence, P5 can end at any time between 16 and 40 minutes. P5 can start at any time between –11 and 13 minutes.
To minimize the time taken for manufacturing, P5 can start at any time between 0 and 13 minutes and can end at any time between 27 and 40 minutes. From the given options, P6, P3 and P2 cannot be the third process to end. Only P5 can be the third process to end.
Choice (A)

7. From the previous solution, if the first process, P6, started at 10:00 AM, P2 will start at 10:19 AM.
Choice (B)

8. We can start with P6 for finding the maximum time required.
P6 can start at 0th minute. It will end at the 33rd minute.
P4 has to start exactly 20 minutes after P6 starts. Hence, P4 must start at 20th minute and end at 34th minute.
P2 must start before P4 ends. P2 can start at 33rd minute and end at 65th minute (if we decrease the starting time, the ending time will also decrease and so, will the overall time taken for these three processes).
P3 has to end exactly 10 minutes before P2 ends. Hence, P3 must end at 55th minute and must start at 36th minute.
P1 must start at most 5 minutes after P3 ends. The latest time at which P1 can start is 60th minute. Hence, P1 can end at the 88th minute.
P5 must end at most 25 minutes before P3 ends. It must end at any time between 30th minute and 54th minute (since P3 ends at 55th minute). Hence, P5 must start at any time between 3rd minute and 27th minute.
This information is tabulated below:

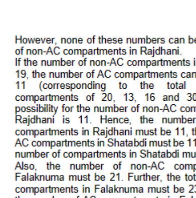
Process	Start Time	End Time
P6	0	33
P4	20	34
P2	33	65
P3	36	55
P1	60	88
P5	3-27	30-54

Hence, the maximum time taken for manufacturing a widget is 88 minutes.
Choice (D)

9. From the previous solution, the first process to start is P6. The second and third processes can be P5 and P4, in any order. The fourth process will be P2 and the fifth process is P3.
Choice (C)

10. Refer to the solution to question 6.
We can see that the minimum time required for manufacturing each widget = 70 minutes.
Choice (B)

Solutions for questions 11 to 15:



Let the letters presented in the adjacent diagrams represent the number of people who play each sport in each group.
From (i), h = j. From (ii), a = 2l.

From (iv), i + k = 95.
Hence, j + l = 120 – 95 = 25
But j = h. ∴ h + l = 25
From (iii), d + h + l = 40 ⇒d = 15
From (v), b + c = 65. Since a + b + c + d = 120, a + 65 + 15 = 120
⇒a = 40
From (ii), l = 20⇒h = 5 = j.
From (v), e + f = 65.
Since e + f + g + h = 120, g = 120 – 65 – 5 = 50
From (iv), a + c = 70 ⇒c = 70 – 40 = 30.
Since b + c = 65 ⇒b = 35
From (iii), b + f + j = 60 ⇒ 35 + f + 5 = 60 ⇒f = 20
Since e + f = 65, **e = 45.**

11. The number of Americans who do not play either of the two sports (d) is 15.
Choice (A)

12. The number of Brazilians who play only Hockey (g) is 50.
Ans: (50)

13. The number of Brazilians who play both the sports (f = 20) is the same as the number of Chinese who do not play either of the two sports (l = 20).
Choice (A)

14. The maximum number of people in a single group who do not play either of the two sports is 20 (for Chinese).
Ans: (20)

15. The number of Chinese who play only Hockey can be a maximum of 95 (since i + k = 95). Hence, it cannot be twice the number of Brazilians who play only Hockey (2 × 50).
Choice (D)

Solutions for questions 16 to 20:

From (iv), the total number of compartments in Rajdhani and Duronto is greater than 37. Also, the total number of non-AC compartments in Falaknuma and Shatabdi must also be greater than 37.
The maximum number of non-AC compartments in Falaknuma and Shatabdi is 19 + 21 = 40 and the next highest is 18 + 21 = 39. No other combination of numbers is possible for the number of non-AC compartments in Falaknuma and Shatabdi.
If the number of non-AC compartments in Falaknuma and Shatabdi is 39, then the total number of compartments in Rajdhani and Duronto must also be 39. However, with the numbers 10, 14, 17, 20, 23, 26 and 30, no two numbers will add up to 39. Hence, the number of non-AC compartments in Falaknuma and Shatabdi cannot be 39 and must be 40.
In this case, the number of compartments in Rajdhani and Duronto can be 23 and 17 OR 26 and 14 OR 30 and 10, in any order.
From (i), the number of AC compartments in Shatabdi is the same as the number of non-AC compartments in Rajdhani. We know that the number of non-AC compartments in Shatabdi must be 21 or 19. If the number of non-AC compartments in Shatabdi is 21, the number of AC compartments can be 2 or 5 or 9 (as the total number of compartments can only be 23 or 26 or 30). However, none of these numbers can be the number of non-AC compartments in Rajdhani.
If the number of non-AC compartments in Shatabdi is 19, the number of AC compartments can be 1, 4, 7 or 11 (corresponding to the total number of compartments of 20, 13, 16 and 30). The only possibility for the number of non-AC compartments in Rajdhani is 11. Hence, the number of non-AC compartments in Rajdhani must be 11, the number of AC compartments in Shatabdi must be 11 and the total number of compartments in Shatabdi must be 30.
Also, the number of non-AC compartments in Falaknuma must be 21. Further, the total number of compartments in Falaknuma must be 23 OR 26 and the number of AC compartments in Falaknuma must be 2 or 5.
From (ii), the number of non-AC compartments in Grand Trunk must be 8 or 11. However, this cannot be 11 as the number of non-AC compartments in Rajdhani is 11. Hence, the number of non-AC compartments in Grand Trunk is 8, the number of AC compartments in Falaknuma is 2 and the total number of compartments in Falaknuma is 23.
We have seen above that the number of compartments in Rajdhani and Duronto can be 23 and 17 OR 26 and 14 OR 30 and 10. However, the first case is not possible as the total number of compartments in Falaknuma is 23 and the last case is not possible as the total number of compartments in Shatabdi is 30. Hence, the only possibility for the total number of compartments in Rajdhani and Duronto is 26 and 14 in any order.
The number of AC compartments in Rajdhani can be 3 or 15. However, from (iii), this must be greater than 5. Hence, the number of AC compartments in Rajdhani must be 15 and the total number of compartments in Rajdhani and Duronto must be 26 and 14 respectively.
The number of non-AC compartments in Duronto must be 6 or 16 or 18. However, since the total number of compartments in Duronto is only 14, the number of non-AC compartments must be 6. Hence, the number of AC compartments in Duronto must be 8.
The number of non-AC compartments in Sampparkranti and Deodhar must be 16 and 18 in any order. Hence, the total number of compartments in any of these trains cannot be 10. Therefore, the only train which can have the total number of compartments as 10 is Grand Trunk.
Between Samparkranti and Deodhar, one train has a total of 17 compartments, with 16 non-AC compartments, while the other has a total of 20 compartments, with 18 non-AC compartments. The former is the train with the least number of AC compartments. Hence, this cannot be Samparkranti.
∴ Deodhar has a total of 17 compartments, with 16 non-AC compartments, while Samparkranti has a total of 20 compartments, with 18 non-AC compartments.
The following table provides, for each train, the total number of compartments, the number of AC compartments, and the number of non-AC compartments:

Train	Total	Non-AC	AC
Shatabdi	30	19	11
Falaknuma	23	21	2
Rajdhani	26	11	15
Grand Trunk	10	8	2
Samparkranti	20	18	2
Deodhar	17	16	1
Duronto	14	6	8

16. Rajdhani has 26 compartments.
Ans: (26)

17. Duronto has the least number of non-AC compartments.
Choice (C)

18. The number of AC compartments in Duronto is equal to the number of non-AC compartments in Grand Trunk.
Hence, the required difference = 0.
Ans: (0)

19. From the above table, we can observe that the minimum seating capacity will be for either Grand Trunk or Duronto.
The seating capacity of Grand Trunk
= 8 × 90 + 2 × 25 = 770
The seating capacity of Duronto
= 6 × 90 + 8 × 25 = 740
Hence, the minimum seating capacity of any train is 740.
Choice (B)

20. In only 3 trains – Shatabdi, Rajdhani and Duronto, the number of AC compartments is more than 5.
Choice (C)

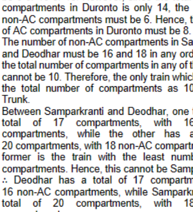
Difficulty level wise summary - Section II

Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	1, 2, 3, 4, 5, 11, 12, 13, 14, 15
Difficult	6, 7, 8, 9, 10, 16, 17, 18, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. The maximum possible area of an isoceles triangle which lies entirely within the square of side 24 cm
$= \frac{1}{2}(24)(24) = 288$ sq. cm Hence, P must above AB
(i.e., P cannot lie inside the square, because it is given that the area common to the square and the triangle is 360 sq.cm., which is greater than 288 sq.cm).



Let PC and PD intersect AB at N and M respectively as shown in the figure above.
Area of the trapezium MNCD $= \frac{1}{2}$ (MN + CD) (BC)
= 360
$= \frac{1}{2}(24 + MN)\, 24 = 360$
⇒ MN = 6
Let us drop a perpendicular from P to CD such that it intersects AB at X and CD at Y
Now, $\frac{PX}{PY} = \frac{MN}{CD}$ [Basic proportionality theorem]
$\frac{PX}{PX + 24} = \frac{6}{24}$
⇒ PX = 8
∴ Area of ΔPCD $= \frac{1}{2}$ (CD) (PY) $= \frac{1}{2}(24)(32)$
= 384 sq units.
Ans: (384)

2. A ≤ 9 and B ≤ 9. Hence A + B ≤ 18
As A + B and $(A + B)^K$ always have the same units digit, A + B must end with 0 or 1 or 5 or 6.
As (A × B) and $(A \times B)^K$ always have the same units digit, A × B must end with 0 or 1 or 5 or 6.
As A + B ≤ 18, A + B can be 5 or 6 or 10 or 11 or 15 or 16 to satisfy the given conditions. We have the following results for each of these values of A + B.
(1) A + B = 5
Possible values of (A, B) are (4, 1) and (3, 2). However, only when (A, B) is (3, 2) the condition for A × B is satisfied.
(2) A + B = 6
Possible values of (A, B) are (5, 1) (4, 2) and (3, 3).
Only when (A, B) is (5, 1) the condition for A × B is satisfied.
(3) A + B = 10
Possible values of (A, B) are (9, 1), (8, 2), (7, 3), (6, 4) and (5, 5). Only when (A, B) is (8, 2), (7, 3) or (5, 5), the condition for A × B is satisfied.
(4) A + B = 11
Possible values of (A, B) are (9, 2), (8, 3), (7, 4) and (6, 5). Only when (A, B) is (6, 5) the condition for A × B is satisfied.
(5) A + B = 15
Possible values of (A, B) are (9, 6) and (8, 7). Only when (A, B) is (8, 7) the condition for A × B is satisfied.

(6) A + B = 16
Possible values of (A, B) are (9, 7) and (8, 8). For neither of these possibilities is the condition for A × B satisfied.
∴ A total of 7 possibilities exist for (A, B)
Ans: (7)

3. $x^3 - 2(x + 2)^2 = 993$
$\Rightarrow x^3 + 2^3 - 2(x + 2)^2 = 1001$
$\Rightarrow (x + 2)\,[x^2 - 2x + 4 - 2x - 4] = 1001$
$\Rightarrow (x + 2)\,(x)\,(x - 4) = 13 \times 11 \times 7 = 1001$
∴ x = 11
Ans: (11)

4. The line parallel to 2x + 3y = 6, will be of the form 2x + 3y = k.
Since 101……… the y-intercept of 2x + 3y = k is given as 5, the point (0, 5) lies on 2x + 3y = k.
Hence, 2 (0) + 3 (5) = k, i.e., k = 15
∴ required equation is 2x + 3y = 15 Choice (C)

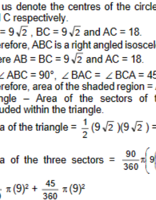
5. Let the three-digit number be abc.
Given that $\frac{a}{b} = \frac{b}{c}$
i.e., $b^2 = ac$
This is possible when a = b = c
∴ we have 9 numbers 111,222 . . . . 999.
Also checking for other values of a, b, c
b = 2, ac = 4 ⇒ abc = 124 (or) 421
b = 3, ac = 9 ⇒ abc = 139 (or) 931
b = 4, ac = 16 ⇒ abc = 248 (or) 842
b = 5, ac = 25 ⇒ No value
b = 6, ac = 36 ⇒ abc = 469 (or) 964
For other values of b, abc does not exist such that digits are distinct.
∴ 9 + 8 = 17 numbers exist
Ans: (17)

6. From 15/08/1947 to 15/08/1997, there are 13 leap years and 37 non-leap years.
Hence, we have a total of 13 (2) + 37 odd days = 63 odd days.
Hence, 15/08/1947 was a Friday.
Choice (D)

7. Given $\frac{a + 10d}{a + 7d} = \frac{2}{5}$
⇒ 5a + 50d = 2a + 14d
⇒ 3a + 36d = 0 → (1)
⇒ a + 12d = 0
⇒ 13th term is zero.
⇒ choice (C) is true.
3a + 36d = (a + 11d) + (a + 12d) + (a + 13d)
0 = sum of three consecutive terms.
∴ choice (B) is definitely false.
The series can be written as –12d, –11d, –10d, ……. d, 0, d, 2d, …….
∴ The sum of no two consecutive terms is zero unless d = 0
∴ Choice (A) is definitely not false.
Choice (B)

8. In the time Anju takes 72 steps, Bobby would take $\frac{72}{K}$ steps. So, the difference in the distance covered by them at this instant will be $72 - \frac{72}{K}$ steps.
Since speed of Bobby and that of the escalator are the same, half of $\left(72 - \frac{72}{K}\right)$ will be covered by Bobby and the remaining half by the escalator, for Bobby to fully climb up the escalator.
Total number of steps taken by Bobby
$= \frac{72}{K} + \frac{1}{2}\left(72 - \frac{72}{K}\right) = \frac{36}{K} + 36$
∴ K must be a factor of 36, except 1, i.e., K can take (9 – 1) = 8 values.
Ans: (8)

9. Joining the centres of the three circles we get the following figure:



Let us denote the centres of the circles by A, B, and C respectively.
AB = $9\sqrt{2}$, BC = $9\sqrt{2}$ and AC = 18.
Therefore, ABC is a right angled isosceles triangle, where AB = BC = $9\sqrt{2}$ and AC = 18.
∴ ∠ABC = 90°, ∠BAC = ∠BCA = 45°.
Therefore, area of the shaded region = Area of the triangle – Area of the sectors of the circles included within the triangle.
Area of the triangle $= \frac{1}{2}(9\sqrt{2})(9\sqrt{2}) = 81$
Area of the three sectors $= \frac{90}{360}\pi\left(9\left(\sqrt{2} - 1\right)\right)^2 + \frac{45}{360}\pi(9)^2 + \frac{45}{360}\pi(9)^2$
$= \frac{1}{4}\pi 81[3 - 2\sqrt{2}] + \frac{1}{8}\pi(81) + \frac{1}{8}\pi(81)$
$= \frac{1}{4}\pi 81[3 - 2\sqrt{2} + 1] = \frac{1}{4}\pi 81\left(4 - 2\sqrt{2}\right)$
$= 81\pi\left(1 - \frac{1}{\sqrt{2}}\right)$
∴ Required area $= 81\left(1 - \pi + \frac{\pi}{\sqrt{2}}\right)$
Choice (B)

10. Initially the volume of water in D_1 is more than the volume of water in D_2 by 60%
Let the capacities of the drums D_1 and D_2 be V_1 and V_2 respectively.
∴ $V_1 = V_2 + \frac{60}{100}(V_2)$
Let V_2 be 5k, therefore V_1 = 8k
Rate at which water leaks from $V_1 = \frac{8k}{6}$ and the rate at which water leaks from $V_2 = \frac{5k}{9}$
Let the time after which the volume of water left in D_2 becomes 25% more than the volume of water left in D_1 be t.
In time t, water leaking out of $D_1 = \frac{8k}{6}t$ and the water leaking out of $D_2 = \frac{5k}{9}t$
Therefore the water left in $D_1 = 8k - \frac{8k}{6}t$ and the water left in $D_2 = 5k - \frac{5k}{9}t$
Now, $\frac{5}{4}\left(8k - \frac{8k}{6}t\right) = \left(5k - \frac{5k}{9}t\right)$
$5 - \frac{5}{9}t = \frac{5}{4}\left(8 - \frac{8}{6}t\right)$
Or, $5 - \frac{5}{9}t = 10 - \frac{5}{3}t$
Or, $\frac{5}{3}t - \frac{5}{9}t = 5$
Or, $\frac{2}{9}t = 1$
Or, t = 4.5 hours.
Choice (B)

11. $3x^3 - 31x^2 + 76x - 22 = 0$.
If $f(x) = ax^3 + bx^2 + cx + d$, then the sum and product of the possible values of x satisfying are given by
$S = -\frac{b}{a} = \frac{31}{3}$ and $P = -\frac{d}{a} = \frac{22}{3}$.
∴ $S - P = \frac{31}{3} - \frac{22}{3} = \frac{9}{3} = 3$.
Choice (A)

12. Let the present ages of Raju and his son be R and S respectively.
2n years ago,
(R – 2n) = 4(S – 2n)
⇒ R = 4S – 6n → (1)
n years ago,
(R – n) = 3(S – n)
⇒ R = 3S – 2n → (2)
(1) = (2)
⇒ 4S – 6n = 3S – 2n
⇒ S = 4n
⇒ R = 3(4n) – 2n = 10n
n years later, total age
= (R + n) + (S + n) = 11n + 5n
= 16n = 80 (given) ⇒ n = 5
Difference in the ages = 10n – 4n = 6n = 30.
Choice (D)

13. In normal pack of cards there are 26 black cards. Two cards can be selected from 26 black cards in $^{26}C_2$ or 325 ways.
But a black card can be drawn in 300 ways.
∴ The missing card is black.
In a normal pack, there are 36 numbered cards. Two cards can be drawn from 36 cards in $^{36}C_2$ or 630 ways.
∴ The missing card is not a numbered card.
Hence, the missing card is a black honour. It can be the king of clubs or the ace of spades. Option (D) is the correct answer.
Choice (D)

14. For a number P, in any base n, the rule for divisibility of P by (n – 1) is that the sum of the digits of P should be divisible by (n – 1) [for example for base 10, we have the rule of divisibility by 9]. Similarly, the rule of divisibility by (n + 1) is that the sum of the alternate digits be a multiple of (n + 1) [for example, in base 10, we have the rule for divisibility by 11]. Hence in the question $(N)_{16}$ must be divisible by both (16 – 1) [since sum of digits = 30 is divisible by 15] and (16 + 1) [since sum of alternate digits is the same]. Hence $(N)_{16}$ must be divisible by both 15 and 17. i.e., $(N)_{16}$ is divisible by $(255)_{10}$.
Choice (C)

15. Let the principal be P and the rate of interest be r% per annum.
Now, $P + \frac{5Pr}{100} = 2P$
$\frac{5Pr}{100} = P \Rightarrow r = 20\%$
Now rate of interest = 20% + 5% = 25%
Let the sum quadruple in n years at 25% per annum simple interest.
$P + \frac{Pn25}{100} = 4P$
$\frac{Pn}{4} = 3P \Rightarrow n = 12$ years.
Choice (A)

16. 300 ml of 40% milk contains 60% × 300 ml of water i.e.,
180 ml water and 120 ml milk.
700 ml of 18% milk contains 574 ml water and 126 ml milk.
Ratio of water and milk in the new mixture
= 754 : 246
= 377 : 123
Choice (B)

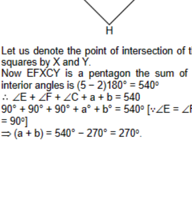
17. Since the answer key of each student is unique (and the total possible answer keys = 5^5 = 3125, which is greater than the total number of students) the minimum number of students getting a positive score = Total number of students – Maximum number of students who can get a zero or a negative score.
To get a negative score, a student should answer all the questions incorrectly, which can be done in 4^5 ways, which, in turn, will be the maximum number of students who can get a net negative score.

To get a zero score, a student has to answer exactly one question correctly and answer all the remaining four questions incorrectly, which can be done in 5 × 4^4 ways, which, in turn, will be the maximum number of students who can get a net zero score.
∴ The minimum number of students getting a net positive score = 3000 – 4^5 – 5 × 4^4
= 3000 – 1024 – 1280 = 696
Ans: (696)

18. 51 + 52 + . . . + N = 5985.
No. of terms in the above series is N – 50.
∴ $(N - 50)\,\frac{(51 + N)}{2} = 5985$
⇒ (N – 50) (N + 51) = 11970
⇒ N^2 + N = 14520.
Ans: (14520)

19. S_o and S_e are odd. Therefore, the number of odd numbers in each of S_o and S_e must be odd, and hence total number of odd numbers = odd + odd = even. We cannot conclude anything about the parity of the number of even numbers and hence nothing about the total number of terms (statements I and II) nor about statement III.
∴ Only IV is necessarily true.
Choice (D)

20.



Let us denote the point of intersection of the two squares by X and Y.
Now EFXCY is a pentagon the sum of whose interior angles is (5 – 2)180° = 540°
∴ ∠E + ∠F + ∠C + a + b = 540
90° + 90° + 90° + a° + b° = 540° [∵∠E = ∠F = ∠C = 90°]
⇒ (a + b) = 540° – 270° = 270°.

Alternative Solution:

With some imagination, we can consider the situation where the second square (EFGH) is rotated, say, counter-clockwise, such that GH becomes parallel to BC (and so will the other corresponding sides). Then b = angle between FG and BC = 90° and a = angle between EH and DC = 180°. Hence, a + b = 90° + 180° = 270°.
Ans: (270)

21. The milkman sells 1ℓ of water + 1.5ℓ of milk = 2.5ℓ of the mixture for ₹(2.5 × 18) = ₹45.
But his effective cost price for the 2.5ℓ of mixture is simply the price of the milk, i.e., ₹(1.5 × 18) = ₹27.
Hence, he makes a profit of ₹(45 – 27) = ₹18 on every ₹27 of his investment.
Hence, the profit percentage that he makes
$= \frac{18}{27} \times 100 = 66.67\%$
Choice (C)

22. $\frac{x^2 - \frac{2x}{3} + 1}{ax^2 + bx + 1} > 0 \Rightarrow \frac{\left(x - \frac{1}{3}\right)^2 + \frac{8}{9}}{ax^2 + bx + 1} > 0$
As $\left(x - \frac{1}{3}\right)^2 + \frac{8}{9}$ is positive, $ax^2 + bx + 1 > 0$
The inequality with the solution set x < –2 and $x > -\frac{1}{2}$ is $(x + 2)\left(x + \frac{1}{2}\right) > 0 \Rightarrow x^2 + \frac{5}{2}x + 1 > 0$
⇒ $ax^2 + bx + 1 > 0$ and $x^2 + \frac{5}{2}x + 1 > 0$ have the same solution ⇒ a = 1 and b = $\frac{5}{2}$
∴ a + b = 3.5
Choice (C)

Difficulty level wise summary - Section III

Level of Difficulty	Questions
Very Easy	–
Easy	4, 6, 15, 16, 18, 21
Medium	7, 10, 11, 12, 13, 19, 20
Difficult	1, 2, 3, 5, 8, 9, 14, 17, 22
Very Difficult	

Done
Tip: Scroll slowly for the best results